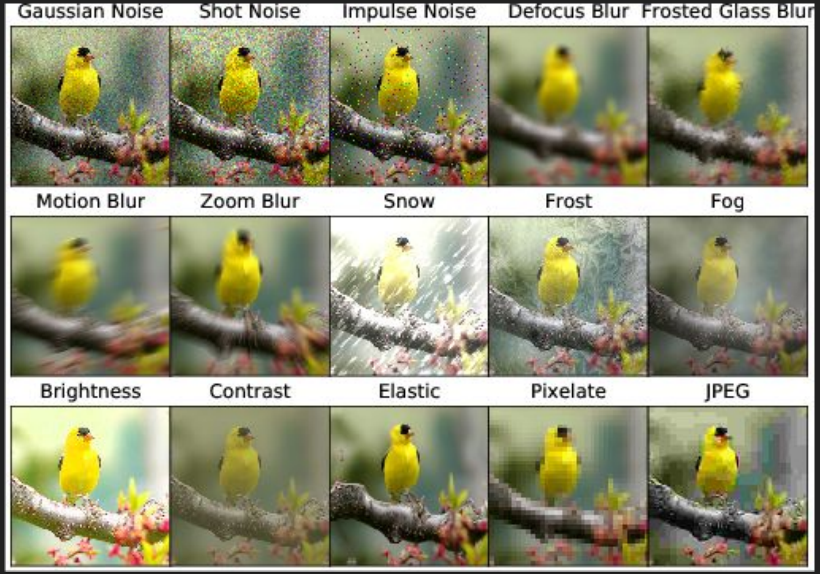
Robustness to Data Distribution Shifts: Test-Time Adaptation Using TENT

Presented by Daniel Gulvin, CSE 710, 11/19/2025

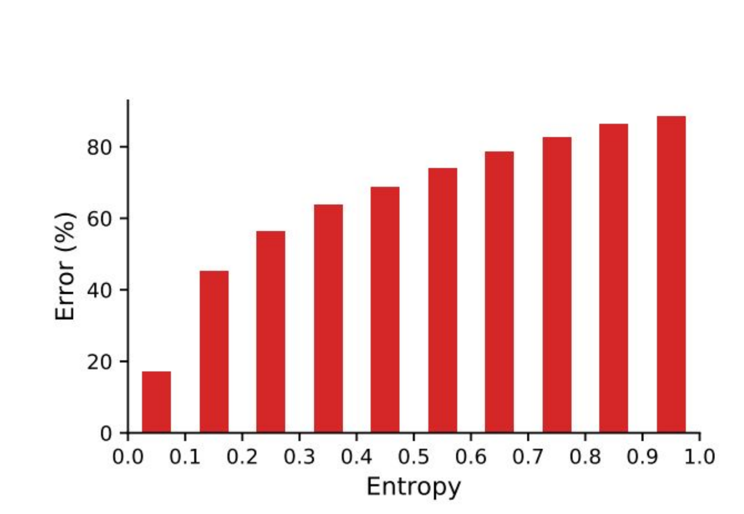
Problem Statement

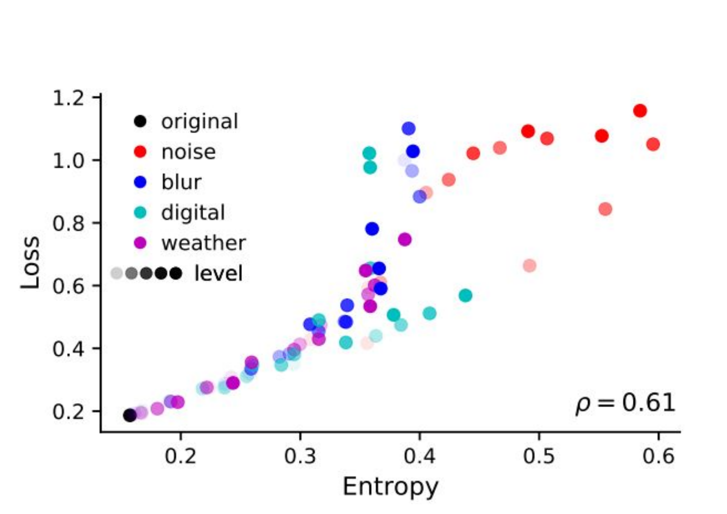
- Distribution shift: test data distribution differs from training
- Retraining is impractical and expensive– requires more labeled data
- **How can we make models more robust to distribution shifts at test time?**



Base Paper - “TENT: FULLY TEST-TIME ADAPTATION BY ENTROPY MINIMIZATION” [1]

- TENT = Test Entropy minimization
  - Allows models to *adapt at test time* by minimizing prediction entropy
- Entropy = uncertainty in model predictions
  - Minimizing entropy makes outputs more confident and reliable under distribution shifts





ImageNet corruption examples (severity 3)

Gaussian noise

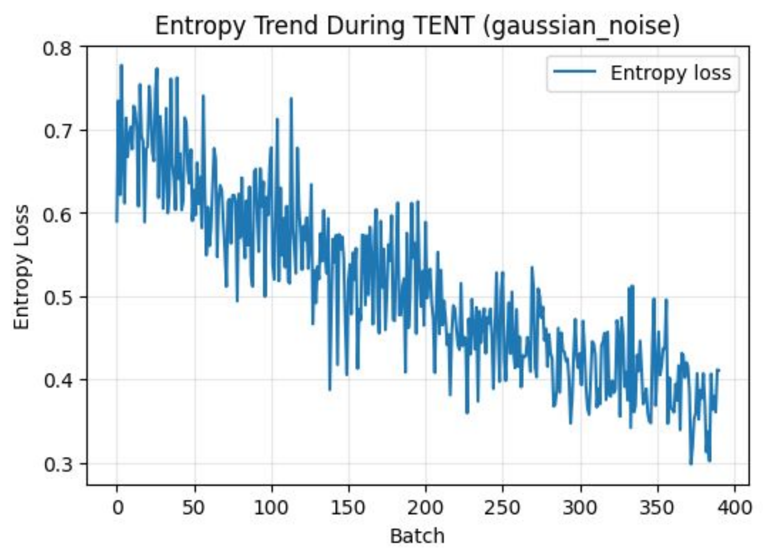
Motion blur

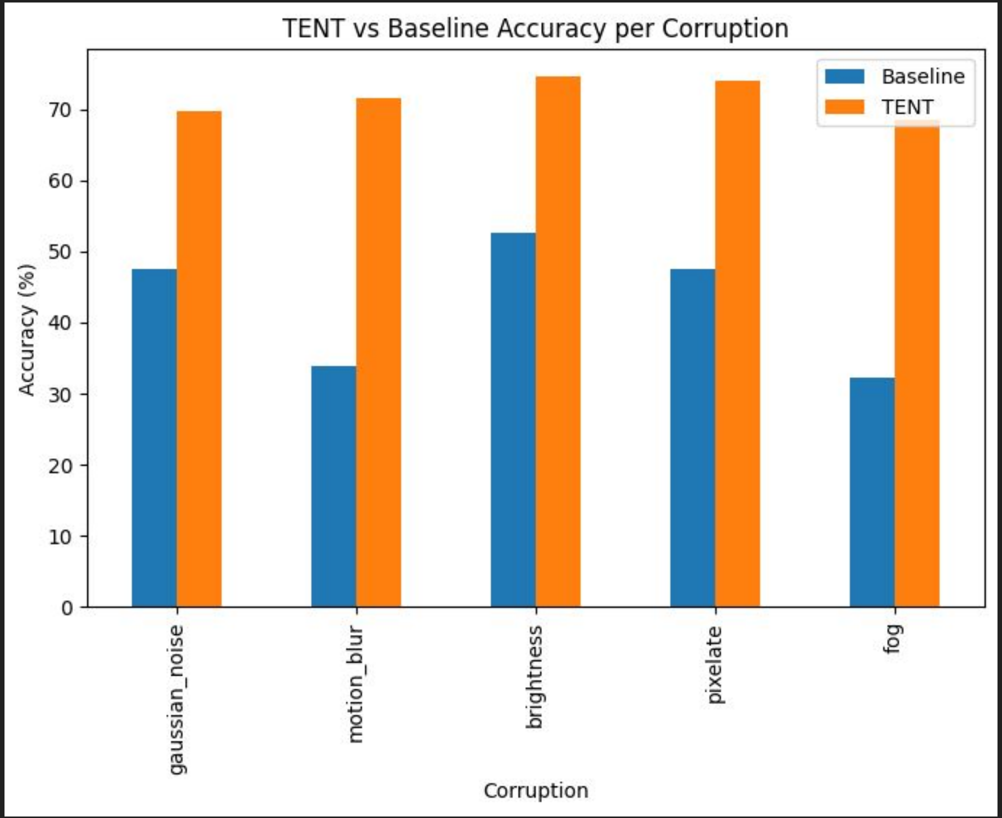
Frost

Experiment #1 - Single Corruption, 10 classes

- Model: ResNet-18
    - Pretrained on ImageNet
    - +1 epoch of fine-tuning final layer on clean CIFAR-10
- Datasets: CIFAR-10/CIFAR-10-C
- Tested:
    - 5 single corruptions (severity 3)
        - gaussian noise, motion blur, brightness, pixelate, fog
- Tracked:
    - Accuracy before and after TENT adaptation
    - Entropy trend (max with 10 classes is 2.3)

Experiment #1 Results

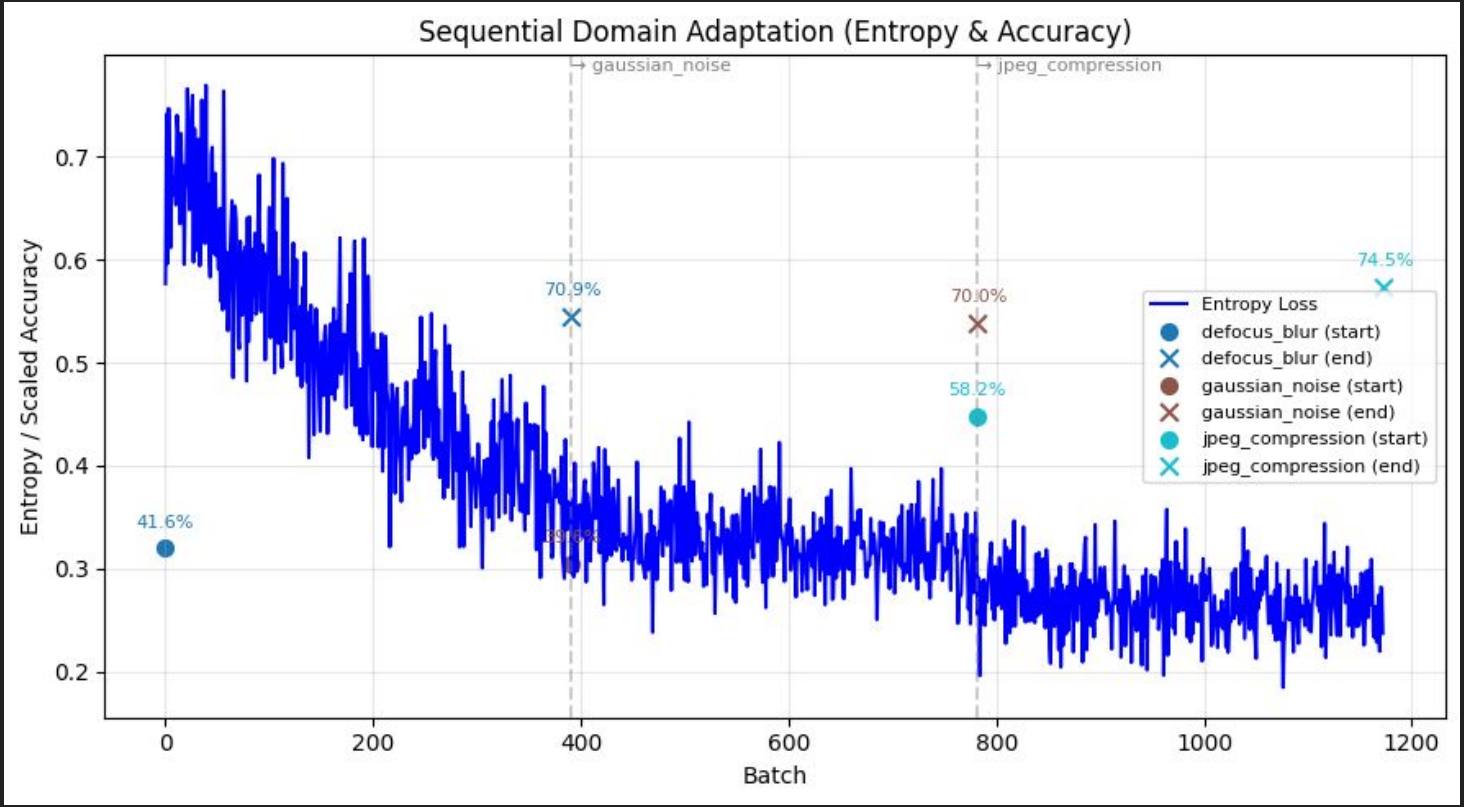




Experiment #2 - Sequential Corruption, 10 classes

- Model: ResNet-18
    - Pretrained on ImageNet
    - +1 epoch of fine-tuning final layer on clean CIFAR-10
- Datasets: CIFAR-10/CIFAR-10-C
- Tested:
    - 3 sequential corruptions (severity 3):
        - Winter weather: frost → snow → fog
        - Rainy weather: brightness → fog → spatter
        - Camera degradation: defocus blur → gaussian noise → jpeg compression
- Tracked:
    - Accuracy before and after TENT adaptation across corruptions
    - Entropy trend

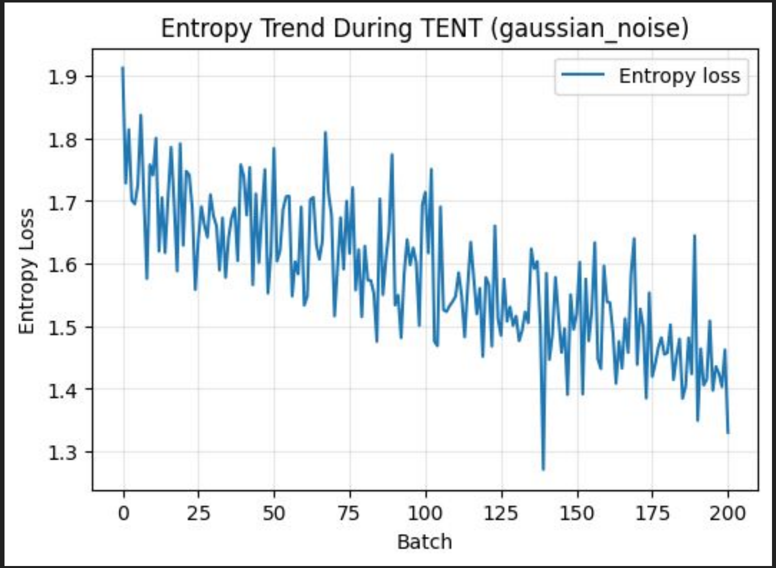
Experiment #2 Results

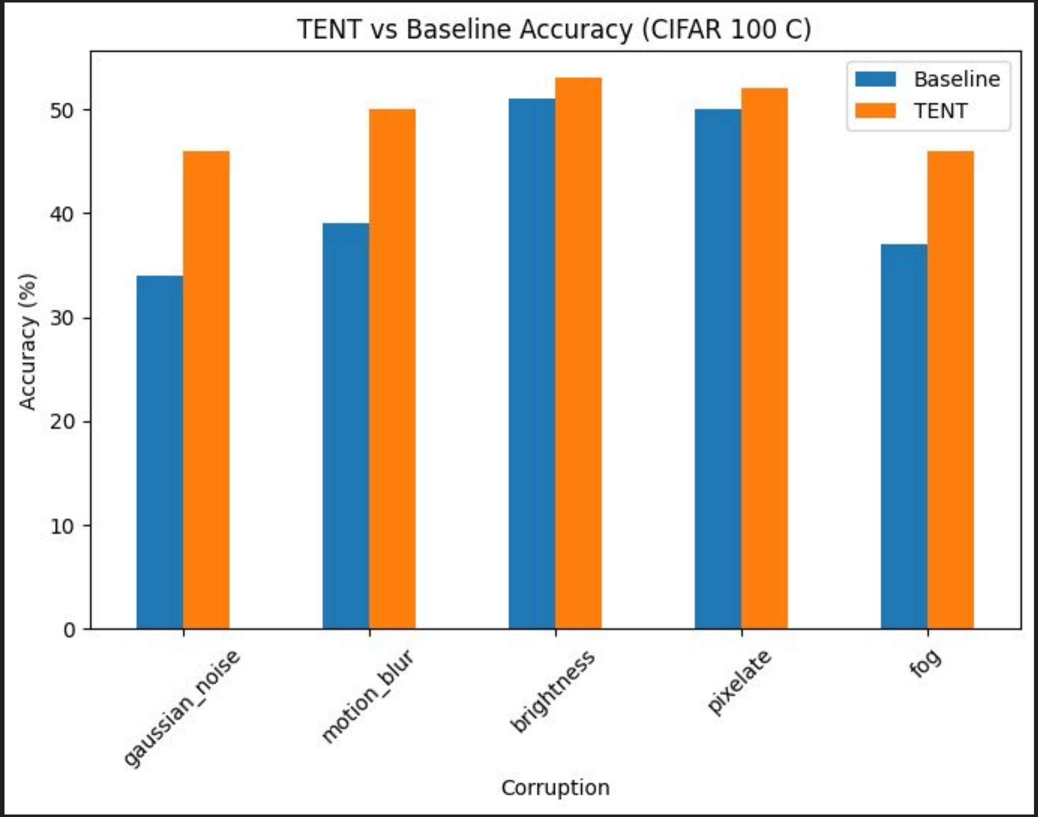


Experiment #3 - Single Corruption, 100 Classes

- Model: ResNet-18
  - Pretrained on ImageNet
  - +10 epochs of fine-tuning final layer on clean CIFAR-100
- Datasets: CIFAR-100/CIFAR-100-C
- Tested:
  - 5 single corruptions (severity 3):
  - gaussian noise, motion blur, brightness, pixelate, fog
- Tracked:
  - Accuracy before and after TENT adaptation (1000 samples)
  - Entropy trend (max with 100 classes is 4.6)

Experiment #3 Results

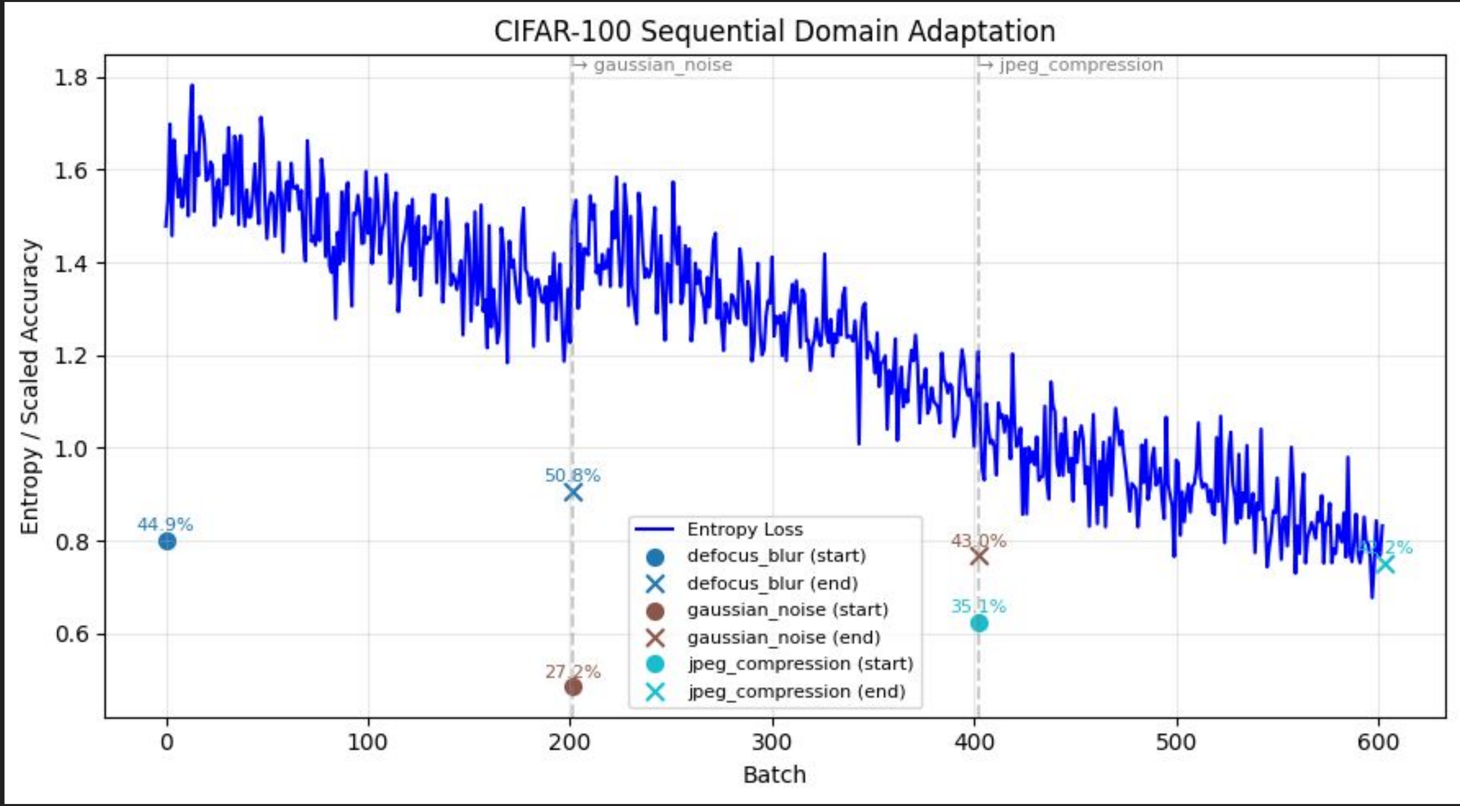




Experiment #4 - Sequential Corruption, 100 classes

- Model: ResNet-18
  - Pretrained on ImageNet
  - +10 epochs of fine-tuning final layer on clean CIFAR-100
- Datasets: CIFAR-100/CIFAR-100-C
- Tested:
  - 3 sequential corruptions (severity 3):
    - Winter weather: frost → snow → fog
    - Rainy weather: brightness → fog → spatter
    - Camera degradation: defocus blur → gaussian noise → jpeg compression
- Tracked:
  - Accuracy before and after TENT adaptation across corruptions (1000 samples)
  - Entropy trend

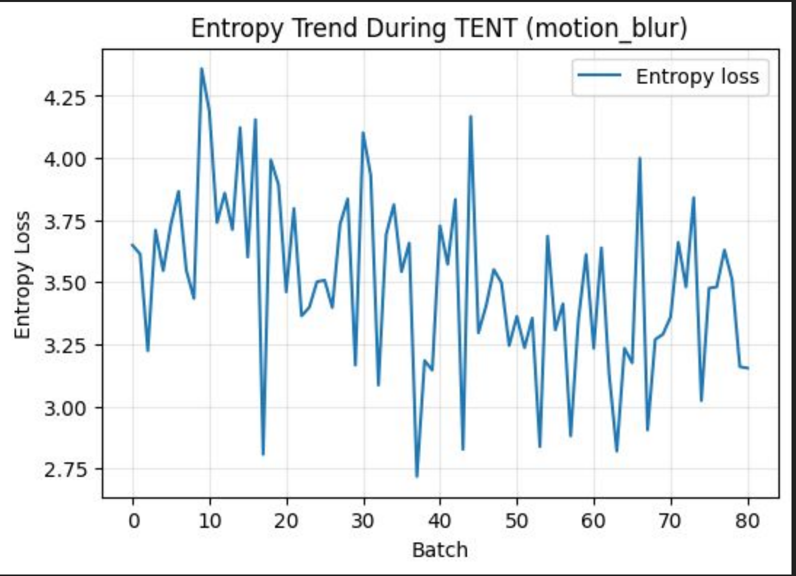
Experiment #4 Results

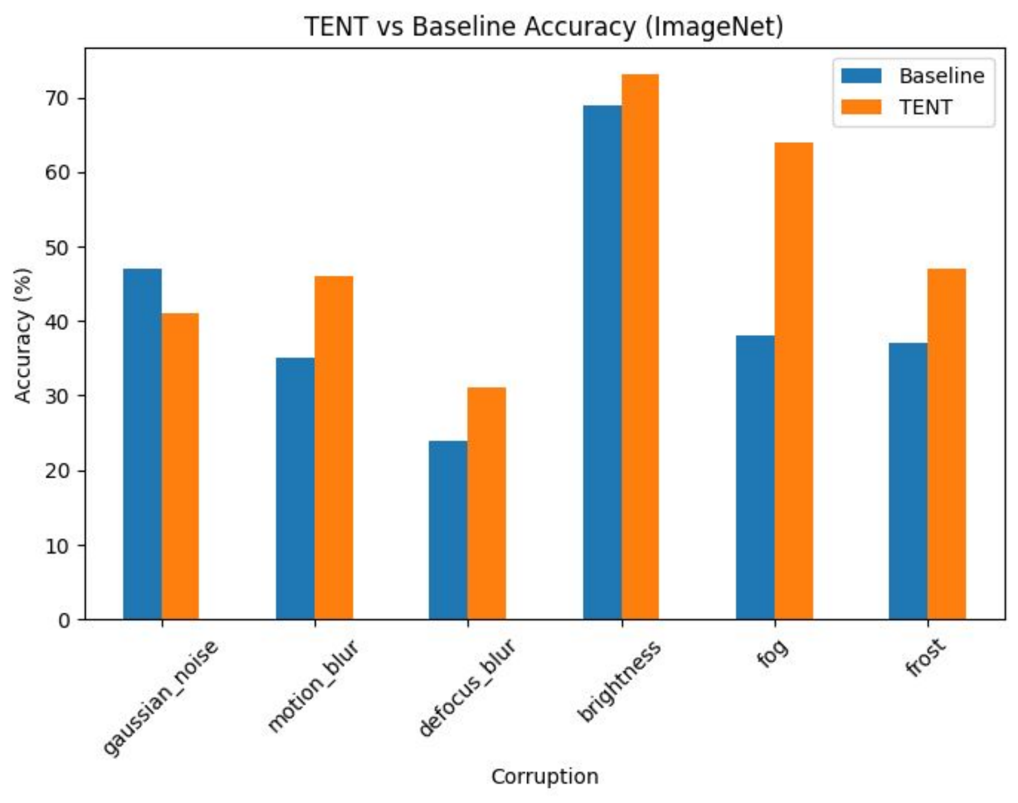


Experiment #5 - Single Corruption, 1000 Classes

- Model: ResNet-50
  - Pretrained on ImageNet
- Datasets: Imagenet/ImageNet-C
- Tested:
  - 6 single corruptions (severity 3):
  - gaussian noise, motion blur, defocus blur, brightness, fog, frost
- Tracked:
  - Accuracy before and after TENT adaptation (2000 samples)
  - Entropy trend (max with 1000 classes is 6.9)

Experiment #5 Results





Takeaways

- TENT consistently improved accuracy across corruptions, but had smaller improvements with more complicated classification.
- TENT struggled the most with noise, likely due to destruction of underlying patterns. Easier corruptions (like brightness) had smaller improvements since they had stronger baselines.
- Entropy steadily decreased during adaptation, but less so for the more complicated classification tasks.
- Sequential adaptation works, but improvements generally lessened after consecutive corruptions. Too many TENT steps could cause overfitting to one corruption, hurting subsequent accuracy with others.